DIAEDDIN RIMAWI

Assistant Professor | Software Engineering, Secure & Resilient AI-Enabled Systems, Process-aware Multi-modal Learning

Bolzano, Italy | diaeddin.rimawi@unibz.it | Personal website | ORCID | Google Scholar | DBLP

ACADEMIC PROFILE

Assistant Professor (RTDa) at the Free University of Bozen-Bolzano (unibz), holding a PhD in Advanced-Systems Engineering, an MSc in Software Engineering, and a Bachelor of Computer Science. My research focuses on engineering secure and resilient AI-enabled systems, with interests in symbolic learning, runtime monitoring, adaptation, and vulnerability intelligence. I teach across undergraduate, master's, and doctoral programs, covering computer vision, software engineering, software quality, and cybersecurity. I bring over a decade of combined experience in university teaching and research, industrial software engineering, and professional technical training, connecting academic knowledge with practical software development and security challenges.

EDUCATION

2022–2025 **PhD in Advanced-Systems Engineering — Free University of Bozen-Bolzano, Italy**
Thesis: Green Resilience of Smart Cyber-Physical Systems. Supervisors: Barbara Russo and Antonio Liotta.

2015–2020 **MSc in Software Engineering — Birzeit University, Palestine**
Thesis: Static and model-based detection of design-pattern use in Android applications.

2010–2014 **BSc in Computer Science — Birzeit University, Palestine**
Thesis: Hybrid intrusion detection system.

ACADEMIC AND INSTRUCTIONAL APPOINTMENTS

Nov 2025–present **Assistant Professor (RTDa) — Free University of Bozen-Bolzano, Italy**
Researcher on the FSE+ funded D2PAM project (multimodal procedural activity understanding and symbolic process mining).
Teaching bachelor's, master's and doctoral courses; design laboratories and assessments; and academic tutor for 15 MSc students (2026/27).

Jan–Nov 2025 **Cybersecurity Technologist — Cybersecurity Laboratory (CSLab) — Free University of Bozen-Bolzano, Italy**
Worked on Cyber Range acquisition, OT/IT security and threat modelling, applied research and proposal development.

Jan 2022–May 2025 **Doctoral Researcher — Free University of Bozen-Bolzano, Italy**
Studied optimization, game-theoretic and reinforcement-learning policies for resilience and sustainability in collaborative AI and cyber-physical systems.

Jun–Dec 2023 **Research Intern — Fraunhofer Italia Research, Italy**
Re-engineered a collaborative-robotic system in-house, for Docker-based execution and evaluated resilience decisions.

Sep 2020–Aug 2022 **Part-time University Instructor — Birzeit University, Palestine**
Taught undergraduate Discrete Mathematics, Database Management Systems and Linux laboratories; supervised undergraduate software projects.

Nov 2020–Dec 2021 **Head of Instructors — AXSOS Academy, Palestine**
Coordinated instructors and teaching assistants in intensive full-stack development bootcamps; previously worked as an instructor using a flipped-classroom, problem-driven approach.

Jul 2017–Nov 2020 **Software Engineer — ASAL Technologies, NVIDIA/Mellanox Firmware Tools team, Palestine**
C/C++/Python software development, debugging, verification, regression-test automation and contributions to open-source MSTFlint components (github.com/diaedinrim contributions).

Sep 2014–Aug 2017 **Teaching and Research Assistant — Birzeit University, Palestine**
Supported introductory computing, C programming and Java data-structures laboratories, student assessment and applied research.

TEACHING EXPERIENCE AND CURRICULUM DEVELOPMENT

University teaching

Introduction to Computer Vision (undergraduate, unibz) — 30 h lectures + 30 h laboratories; image processing, geometry, reconstruction, machine learning and visual understanding.

Software and Systems Security (MSc, unibz) — 20 h laboratory; authorized testbeds, vulnerability investigation, system attacks/defence and evidence-based reporting.

Software Quality and Metrics (MSc, unibz) — 20 h laboratory; GQM-based measurement design and progressive software-quality investigations.

Cybersecurity (PhD, unibz) — 20 h applied teaching in a co-taught course with Prof. Barbara Russo; practical red-team/blue-team and security scenarios.

Database Management Systems (undergraduate, Birzeit) — 60 h lectures; understanding relational databases and apply MySQL queries.

Discrete Mathematics (undergraduate, Birzeit) — 60 h lectures; in-presence and online delivery during COVID-19.

Linux Operating Systems (undergraduate, Birzeit) — 40 h laboratory teaching; in-presence and online delivery during COVID-19.

Introduction to Computers, Introduction to Programming (in C/C++), Data Structures (in Java) (undergraduate, Birzeit) — 20 h each of undergraduate laboratory teaching as Teaching and Research Assistant.

Selected teaching initiatives and evidence

My curriculum development focuses on helping students progressively build their knowledge and apply it to increasingly complex problems. At unibz, I developed connected laboratory sequences in Software and Systems Security and Software Quality and Metrics, where students move from foundational activities toward more independent investigations. I also design practical assignments, written exams, and oral assessments around realistic scenarios, encouraging students to apply concepts, compare approaches, and justify their technical decisions.

Beyond individual courses, I look for opportunities to connect classroom learning with wider engineering activities. For example, I involved undergraduate Computer Vision students in an interdisciplinary drone project associated with the 2026 Conference on Learning Factories. Our course contributed to constructing a 3D model of the student-developed drone, connecting computer vision concepts with work carried out in other engineering courses.

I also maintain publicly available teaching materials and interactive notebooks: github.com/dmrimawi/courses-library

My teaching approach, curriculum development, and student evaluations, including a recent unibz professor score of 9.6/10, can be happily discussed further.

Undergraduate supervision and academic mentoring

Undergraduate team project — Birzeit University: Supervised development of a Flutter campus car-sharing application, integrating pickup-point selection, route efficiency and driver compensation.

Undergraduate security project — unibz CSLab: Supervised students developing an experimental SAST-to-DAST pipeline focused on SQL injection using Docker-based environments and MITRE Caldera.

Academic tutoring — unibz (2026/27): Assigned advisor for 15 MSc Software Engineering students on study planning and academic progression.

SELECTED 10 PEER-REVIEWED PUBLICATIONS

Journal articles

J1 R. Othman, **D. Rimawi**, B. Rossi, and B. Russo. "From Attack Descriptions to Vulnerabilities: A Sentence Transformer-Based Approach." Journal of Systems and Software, 231, 112615 (2026). DOI

J2 **D. Rimawi**, A. Liotta, M. Todescato, and B. Russo. "GResilience: Decision-Making Policies for Trading Off Greenness and Resilience in Online Collaborative AI Systems." IEEE Access, 13, 140867–140888 (2025). DOI

Conference and workshop papers

C1 Z. Razzaq, N. Brahimi, H. Z. U. Rehman, and **D. Rimawi**. "A Comparative Evaluation of RGB-Depth Fusion for Egocentric Assembly Action Recognition." IC2IE (2026); accepted and presented, proceedings publication pending.

C2 A. Janes, **D. Rimawi**, R. T. Othman, A. Avritzer, R. Mirandola, and B. Russo. "Toward Performance-Based Detection of Attacks in Microservice Systems." ICPE Companion (2026), 100–101. DOI

C3 R. Othman, **D. Rimawi**, B. Rossi, and B. Russo. "Predicting Known Vulnerabilities from Attack News: A Transformer-Based Approach." ITASEC & SERICS, CEUR Workshop Proceedings, Vol. 4198 (2026). Paper

C4 **D. Rimawi**, A. Liotta, M. Todescato, and B. Russo. "Modeling Resilience of Collaborative AI Systems." IEEE/ACM CAIN (2024), 24–29. DOI

C5 D. Rimawi, A. Liotta, M. Todescato, and B. Russo. "CAIS-DMA: A Decision-Making Assistant for Collaborative AI Systems." PROFES (2023), 183–199. DOI

C6 **D. Rimawi**, A. Liotta, M. Todescato, and B. Russo. "GResilience: Trading Off Between the Greenness and the Resilience of Collaborative AI Systems." ICTSS (2023), 266–273. DOI

C7 **D. Rimawi**. "Green Resilience of Cyber-Physical Systems." ISSREW (2022), 105–109. DOI

C8 **D. Rimawi** and S. Zein. "A Model-Based Approach for Android Design Patterns Detection." ISMSIT (2019), 1–10. DOI

Further publications can be explored on Google Scholar

RESEARCH PROJECTS, INFRASTRUCTURE AND FUNDING INVOLVEMENT

2025–present	**D2PAM — Data-driven Procedural Activity Mining from Multi-modal Streams** FSE+ funded, Free University of Bozen-Bolzano. **Role**: researcher; multimodal learning, procedural activity understanding and process mining.
2025	**Cyber Range infrastructure — CSLab, unibz** Contributed requirements, design and procurement assessment for teaching/research security infrastructure. **Role**: managed European tender process.
2025	**AI-based security for OT networks (Zero Trust CPS)** European Digital Innovation Hub proposal. **Role**: co-authored technical content and milestones for AI-based intrusion detection and Zero Trust in OT networks. Result: Accepted with 60k euros fund.
2025	**Critical-infrastructure security and NIS2 threat modelling** Public-sector applied-research collaboration. **Role**: co-author/contributor to threat modelling and recommendations. Result: Accepted with 55k euros fund.
2022–2024	**PON Scholarship for Innovation and Green Studies** Doctoral scholarship from the Italian Ministry of University and Research (MUR), supporting dissertation research. **Role**: student.

ACADEMIC SERVICE

Programme committees: CAIN (2026, 2027); ECSA Open Science (2026); MSR4P&S (2026); ICSOB Posters & Demos (2025); SEAA sub-reviewing (2025).

Journal reviewer: IEEE Transactions on Automation Science and Engineering (2026); IEEE Access (2026); International Journal of Machine Learning and Cybernetics (2026); Information and Software Technology (2024–2025).

TECHNICAL EXPERTISE AND QUALIFICATIONS

Security and software engineering: Vulnerability analysis, security testing, threat modelling, Cyber Range infrastructure, OT/IT security, software architecture, software quality/metrics, requirements, static analysis, testing and runtime monitoring.

Methods and platforms: Process mining, reinforcement learning, optimization, computer vision and multimodal AI

Python, C, C++, Java, SQL, Linux scripting, Git, Docker and ROS.

Languages and certificates: Arabic (native), English (C1), Italian (A1.2)

unibz English C1 certificate (2024); Privacy in Research / GDPR training (2024).